

UJIAN AKHIR SEMESTER
Pengolahan Citra dan Pola

Nama : I Putu Evo Wira Saputra

Nim : 2401010250

Jurusan : IF-MTI

Mata Kuliah : Pengolahan Citra dan Pola - C

2401010250_ I Putu Evo Wira Saputra_UAS.py

```
import cv2
import numpy as np

nama_file_input = 'komponen_mesin.jpg'
img = cv2.imread(nama_file_input)

if img is None:
    print(f"Error: File '{nama_file_input}' tidak ditemukan.")
else:
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    blurred = cv2.GaussianBlur(gray, (11, 11), 0)

    _, thresh = cv2.threshold(blurred, 0, 255,
cv2.THRESH_BINARY + cv2.THRESH_OTSU)

    kernel_close = cv2.getStructuringElement(cv2.MORPH_RECT,
(35, 35))
    kernel_open = cv2.getStructuringElement(cv2.MORPH_RECT,
(9, 9))

    morphed = cv2.morphologyEx(thresh, cv2.MORPH_CLOSE,
kernel_close)
    morphed = cv2.morphologyEx(morphed, cv2.MORPH_OPEN,
kernel_open)

    contours, _ = cv2.findContours(morphed,
cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)

    total_objek = 0

    for cnt in contours:
```

```

        area = cv2.contourArea(cnt)
        x, y, w, h = cv2.boundingRect(cnt)

        if x > int(lebar_gbr * 0.12) and y > int(tinggi_gbr *
0.18):

            if area > 2000:
                total_objek += 1

                cv2.rectangle(img, (x, y), (x + w, y + h),
(0, 255, 0), 2)

            teks_informasi = f"Total Objek: {total_objek}"
            posisi_teks = (80, 170) # Diletakkan sedikit di bawah
teks bawaan gambar agar rapi
            font = cv2.FONT_HERSHEY_SIMPLEX
            skala_font = 1.2
            warna_teks = (0, 255, 0)
            ketebalan_teks = 4

            cv2.putText(img, teks_informasi, posisi_teks, font,
skala_font, warna_teks, ketebalan_teks, cv2.LINE_AA)

            nama_file_output = 'hasil_inspeksi.png'
            cv2.imwrite(nama_file_output, img)

            print(f"Sistem Terkalibrasi! Jumlah objek terdeteksi:
{total_objek}")

            # Tampilkan jendela hasil akhir
            cv2.imshow('Hasil Inspeksi', img)
            cv2.waitKey(0)
            cv2.destroyAllWindows()

```

HASIL

hasil_inspeksi.png

